Hyperledger Fabric: An Overview of its Transaction Validation Feature

Hyperledger Fabric, a blockchain implementation hosted by the Linux Foundation, has been the subject of a series of articles in the last few issues of *OSFY*. In this issue, we take a look at how transactions are validated in the blockchain.

During the simulation of a transaction by an endorser, a read-write set is prepared for the transaction. The read set contains a list of unique keys and their committed versions that the transaction reads during simulation. The write set contains a list of unique keys and their new values that the transaction writes, though there can be overlap with the keys present in the read set. A delete marker is set, in the place of a new value for the key if the update performed by the transaction is to delete the key. If the transaction writes a value multiple times for a key, only the last written value is retained. Also, if a transaction reads a value for a key, the value in the committed state is returned even if the transaction has updated the value for the key before issuing the read. In other words, *read-your-writes* semantics are not supported. Or, the versions of the keys are recorded only in the read set; the write set just contains the list of unique keys and their latest values set by the transaction.

There could be various schemes for implementing versions. The minimal requirement for a versioning scheme is to produce non-repeating identifiers for a given key. For instance, using monotonically increasing numbers for versions can be one such scheme. In the current implementation, we use a blockchain height-based versioning scheme in which the height of the committing transaction is used as the latest version for all the keys